



13016223: Artificial Intelligence

Assignment 1: Tennis Ball Machine

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Introduction

Nowadays, technology plays an extremely important role in our daily lives especially the way we live our life tends to be more comfortable and more efficient. For example, we prefer to have a coffee produced by the automatic coffee machine when we are being in the rush hour. We prefer to leave a vacuum-cleaner robot at home when we do not have much time to do our housework by ourselves. All of these agents cannot perform effectively and efficiently without a good artificial intelligence built into them. In this paper, I would like to introduce my favourite agent called a '**Tennis Ball Machine**'. In athletics, as you definitely know that aside from coaching, the only way to get better is to practice and the tennis ball machine is the best partner for you to practice with.

Description

The tennis ball machine is capable of generating shots for those who want to practice like you are playing with your partner. With adjustable speed, spin, ball feed interval, trajectory, and a random oscillator function which means the tennis ball machine shoots balls randomly from side to side. Furthermore, it can detect the position of the player and challenge them by shooting the ball to the other side of the court where it is left no man. I will detail each element of this agent by referring to the **Figure 1** below.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Tennis ball machine	Robust, precise, controllable, safe, movable, consistent	Tennis court, tennis player, spectator	Counter rotating wheel, random oscillator, shot placement, drop, lob, stop, accelerator	Camera, accelerometer, speed sensor, motion sensor, weight sensor, remote control

Figure 1: PEAS description of the task environment for a tennis ball machine

Agent Type: Tennis Ball Machine

Performance Measure

- **Robust:** To ensure that the agent must be able to work even if it is hit severely several times by the balls. (In this case, the agent program itself not the hardware or materials of the machine)
- **Precise:** The agent must shoot the balls precisely as it is programmed.
- **Controllable:** The agent is controllable, in order to be perceived instructions and set the shooting mode, for example, random shots, challenge shots, practice shots, spin shots, etc.
- **Safe:** To maximize safety to the tennis players or the spectators, it should know not to shoot the balls directly to human.
- **Movable:** The agent can move itself to make more challenge to the players.
- **Consistent:** The shot made by the agent should be consistent; that is, the velocity and the acceleration for each shot of the ball should be the same. Excepting, it is set by the player.

Environment

- **Tennis court:** The whole area of the tennis court observed by the agent. (The agent is usually supposed to be positioned at the base line of the court.)
- **Tennis player:** All of the players participate in the court or interact with the agent.
- **Spectator:** For security reason, if the spectator is detected, the agent should automatically know and stop shooting.

Actuators

- **Counter rotating wheels:** In order to shoot the balls, these wheels are needed; especially, to generate the topspin, backspin, and sidespin shots.
- **Random oscillator:** To make the agent to produce shots randomly on anywhere in the court.
- **Shot placement:** To make the agent a precise shot placement when needed.
- **Drop:** To produce drop shots.
- **Lob:** To produce lob shots.
- **Stop:** To stop shooting.
- **Accelerator:** To increase the velocity of the shots.

Task Environment and its characteristics

Task Environment	Observable	Agent	Deterministic	Episodic	Static	Discrete
Tennis ball machine	Fully	Single	Deterministic	Episodic	Semi	Discrete

Figure 2: The task environment and the characteristics of the tennis ball machine

Task Environment: Tennis Ball Machine

Fully observable: The agent's sensors, camera and motion sensor, give it access to the complete state of the environment, the tennis court in this case. Whatever the mode the agent is operating, the shots must be produced only in this court.

Single agent: The agent is clearly in a single agent due to the fact that it is supposed to be a partner of tennis players and it can work by itself.

Deterministic: Since the task environment of the agent is fully observable and the goal of the agent is to shoot the balls, so the environment is deterministic.

Episodic: The shooting modes and each time the agent operate depend on the percept (e.g. instruction by the user), each mode is performed when the agent receives a percept and each mode does not depend on the mode taken in previous modes. Hence it is in episodic task environment.

Semidynamic: Basically, if the player is in the court the environment is static. However, the environment might change while the agent is deliberating. That is, there is no man (the environment, tennis player is absent from agent sight) in the court, the agent will continuously check whether there is a man on the court and make a decision to stop shooting by itself. In this case, the environment is semidynamic.

Discrete:  The machine has a finite set of the balls to be shot, and if it detects that the balls are used up, it will stop working. So, the environment is a discrete-state.